



MPhys Degree Assessments 2024/25

School:

PHYSICS, ENGINEERING AND TECHNOLOGY

Course:

SCIENTIFIC SUPERCOMPUTING — ASSIGNMENT

Allocation of Marks:

This assignment is marked out of 100.

Instructions:

Answer ALL sections.

This is an open assignment. You may refer to notes, textbooks and online information.

You may not discuss this assignment, through direct or indirect means, with any student or any other person until the results and feedback have been released.

Work should be submitted as single archive file in ZIP format – other formats (e.g. tar, RAR, 7zip etc) are NOT acceptable. This archive is to contain a document (either Word DOC or DOCX with embedded fonts or PDF format – other formats e.g. ODT are NOT acceptable) that contains all parts of your solution, including a listing of all your programs, AND separate files containing all the source code, header files, scripts etc. **Failure to comply with these requirements may result in no marks being awarded.**

You must ensure that no identifying information (including your exam number) appears anywhere within your submission. This includes file names, program comments, PDF properties, etc.

Please submit your solution via the Scientific Supercomputing VLE page.

If – and only if – the VLE will not accept your file, you may e-mail it to pet-assessments@york.ac.uk, ensuring that the subject line of your e-mail and the name of the attached file consist of the module code, and the assignment number.

Answer Q1-3 and EITHER Q4 or Q5

- 1 ARCHER2 <https://www.archer2.ac.uk> is the current UK national supercomputer. Bede <https://n8cir.org.uk/docs/bede/> is the current N8 tier-2 supercomputer. These two machines have very different architectures (see the websites for details and also the Bede documentation). Compare and contrast the different hardware, and comment on the design decisions that went into each machine. Which do you think would be best for a biomolecular simulation using Gromacs and why?

[2 sides A4 max]

[15]

- 2 Consider a simple model 2D problem, which has a regular 2D square grid of data with periodic boundary conditions on all edges. Using the same approach and assumptions as in the lectures, perform your own “cost of parallel decomposition analysis”, and hence derive expressions for the figure of merit, $P\Sigma/N_{grid}$, for both 1D and 2D data decompositions. How many processors are required before a 2D decomposition scheme becomes more efficient?

[10]

- 3 Download this tar-gzip file from http://www-users.york.ac.uk/~mijp1/exam_q3.tar.gz and unpack it using

```
gunzip exam_q3.tar.gz
tar -xf exam_q3.tar
```

and then move into the new directory created. This contains a simple sphere packing code, in both Fortran and C++ variants. Decide which language variant you wish to work with, and then compile it using appropriate flags. When executed, the code will produce some output to the screen, and also create an output file called `final-positions.txt`.

- (a) Profile the code and identify the performance bottleneck(s). Explain why you think these sections of the code are slow, and what can be done to accelerate them using OpenMP. Include the appropriate section(s) of the profile output in your answer.

[5]

- (b) Add appropriate OpenMP directives and function calls to improve the performance of the code. Explore different OpenMP directives to achieve the best parallel speedup you can get. Explain what you have done and why, and what actions (if any) were necessary to ensure the code gives the same answer on any number of threads (both output to screen and output to file) as in serial.

[15]

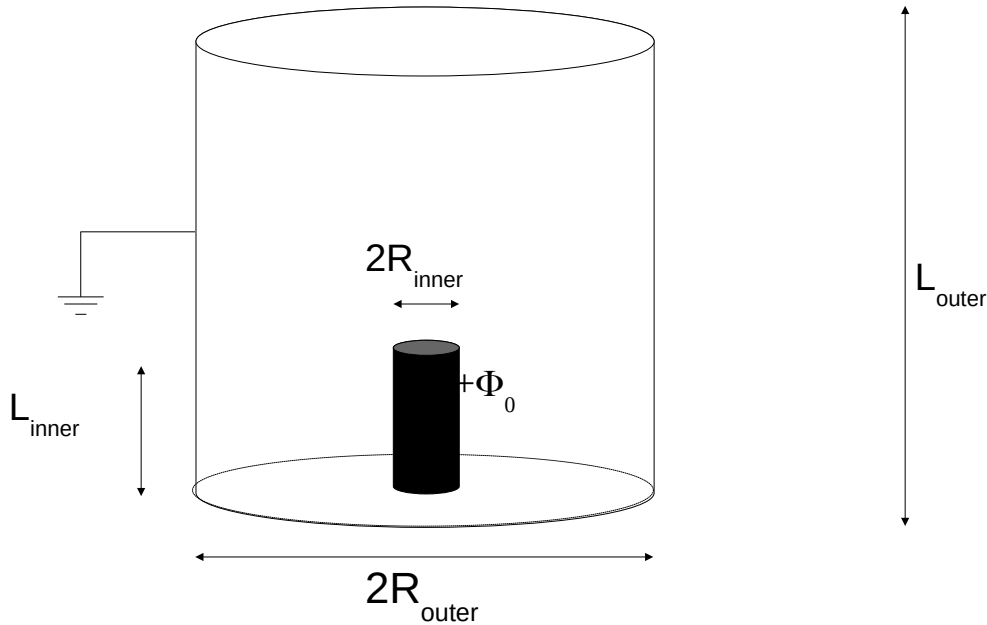
- (c) What is the scaling behaviour of your modified code on `viking`? What speedup can you get with `OMP_NUM_THREADS` varying from 1 to 96? Include a graph of speedup in your answer. Explain the behaviour you observe in terms of the code

and the hardware.

[5]

DO ONE OF EITHER Q4 or Q5

- 4 Consider the following simple electrostatic problem:



A coaxial conductor comprises two concentric conductors: an outer hollow cylindrical metal tube of radius R_{outer} which is earthed on its outer surface, and an inner metal conductor of radius R_{inner} at potential Φ_0 . The end of a long piece of this conductor is shown, where the inner conductor is of length L_{inner} beyond some reference plane, and the outer is of length L_{outer} beyond the same reference plane. This reference plane is chosen such that the potential at that plane is the same as that in the perfect infinitely long coax conductor:

$$\Phi(\phi, r, z) = \frac{\log(R_{outer}) - \log(r)}{\log(R_{outer}) - \log(R_{inner})} \Phi_0$$

using cylindrical polar coordinates.

As there are no free charges, then the electrostatic potential $\Phi(\phi, r, z)$ inside the coax cavity can be found by solving the Laplace equation with appropriate boundary conditions and symmetry. With the rotational symmetry shown, it can be seen that the problem is essentially a two-dimensional one.

The Laplace equation can be solved using the successive over-relaxation (SOR) method:

- Create a grid of points within the region defined by the boundaries
- Use given boundary conditions to assign initial values for the potential, etc.

REPEAT

- For each grid point in turn, calculate a new value for the potential U at that point due to its neighbours
- Use this value U to update the actual potential at that particular grid point using $\Phi^{new} = \Phi^{old} + w(U - \Phi^{old})$ where w is a free parameter ($1 \leq w < 2$) that controls the rate of convergence to the solution.
- NB Whenever a new value of Φ is calculated, the old value at that grid point is *immediately* over-written by the new one – hence only one array of potential values is required. The order in which the points of the grid are updated is arbitrary. [For the curious: this is a Gauss-Seidel update and not the Jacobi update that was used in the weekly problems].

UNTIL the solution has converged, i.e. when $\Phi^{new} = \Phi^{old}$ to an acceptable degree of accuracy.

(a) In cylindrical polar coordinates, the Laplacian is given by

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial z^2} + \frac{1}{r} \frac{\partial \Phi}{\partial r} + \frac{\partial^2 \Phi}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 \Phi}{\partial \theta^2}, \quad r > 0$$

which must be discretised before it can be used on a grid as described above. Using a centred-difference approximation* to each derivative as appropriate, show that an appropriate discretisation of the rotational symmetric Laplace equation in cylindrical polar coordinates is given by

$$\Phi_{i,k} = \frac{1}{4} (\Phi_{i+1,k} + \Phi_{i-1,k} + \Phi_{i,k+1} + \Phi_{i,k-1}) + \frac{1}{8k} (\Phi_{i,k+1} - \Phi_{i,k-1}), \quad r > 0$$

where $z = i * h$ and $r = k * h$, with h being the grid size and i, k being integers.

For $r = 0$, a special case formula (the "axis formula") is required. The difficulty at $r = 0$ in the Laplacian can be circumvented by using Cartesian coordinates. Hence show that an appropriate discretisation of the Laplace equation in cylindrical polar coordinates for a point on the $r = 0$ axis is given by

$$\Phi_{i,0} = \frac{2}{3} \Phi_{i,1} + \frac{1}{6} (\Phi_{i+1,0} + \Phi_{i-1,0}), \quad r = 0$$

The centred-difference formulae for ordinary derivatives are

$$\frac{df}{dx} \approx \frac{f(x+h) - f(x-h)}{2h}$$

and

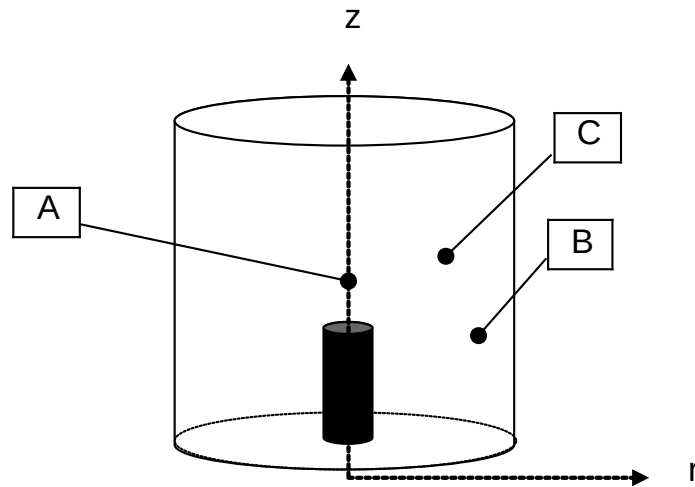
$$\frac{d^2 f}{dx^2} \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

[10]

- (b) Write a computer program that uses this method of successive over-relaxation to solve for the electrostatic potential within the cylindrical region between the outer and inner conductors for any given values of L_{inner} , L_{outer} , R_{inner} and R_{outer} [assuming $L_{inner} < L_{outer}$ and $R_{inner} < R_{outer}$].

Your program may be serial or parallel, and run on either CPU or GPU. More marks will be available for parallel code - see below for details.

- (i) For the particular case that $L_{outer} = 20$ mm, $R_{outer} = 10$ mm, $L_{inner} = 5$ mm, $R_{inner} = 1$ mm and $\Phi = 1000$ V, find the value of the electrostatic potential to mV accuracy (which will require careful convergence), at the following points as shown:



- A: $z = 7.5$ mm, $r = 0$ (i.e. on the central axis)
- B: $z = 5.0$ mm, $r = 6.0$ mm
- C: $z = 12.5$ mm, $r = 5.0$ mm

and include a plot of the potential within the coax cavity. Marks will be awarded for correctness and convergence, as well as for neatness/clarity of coding, and

documentation.

[20]

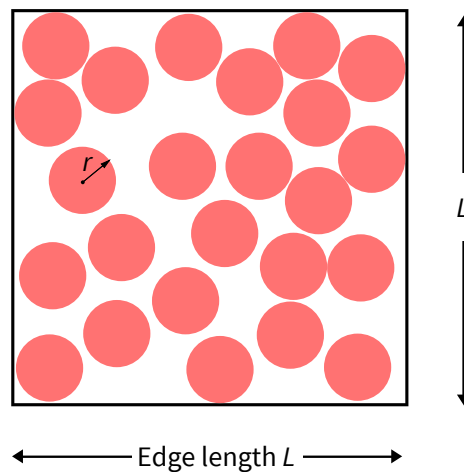
- (ii) Profile your code, and discuss the actions you have taken to make your code as efficient as reasonable. Include one or more appropriate annotated profiles.

[5]

- (iii) Demonstrate how your program scales with different numbers of processing elements (either cores or processors or GPUs etc) and with different density solution grids. Include appropriate graph(s) showing speedup (and optionally run-time as well) and clearly document all relevant parameters. You must include which machine(s) you have used, compilers and flags, etc. and prove that your answers do not change with the number of processing elements.

[15]

- 5 Random packing is often an issue for packaging of goods to correctly estimate the volume or weight of a product within a box. The packing fraction depends on the shapes of the contents and packaging and for random packing the solution is not analytical, requiring a computational model of the packing of objects. Such a model can also be used to approximate the structure of porous materials such as sponges. The packing fraction is often optimised by shaking to reduce the size of voids, but the first stage is to randomly pack the contents in a starting configuration. Consider random 2D packing of uniform circles of radius r with a 2D box with edge length L :



The initial packing is done by generating random x and y coordinates for each circle and attempting to place the circle in the box so that it does not overlap the box boundary or any other circle in the box. This is determined by ensuring the distance between the centre of any two circles is greater than $2r$. New circles are generated until no more circles can be placed, requiring many attempts as the number of possible voids decreases. The final packing fraction P is given by:

$$P = \frac{\pi N_{\text{circ}} r^2}{L^2}$$

where N_{circ} is the number of circles placed in the box, r is the radius of the circles, and

L is the size of the box.

- (a) Write a serial computer program to calculate the maximum packing fraction P as defined in the equation above for a box size $L = 500$ and circle radius $r = 1.234$. Show the convergence of the packing fraction P as a function of the number of trial circle placements and give your final value of P in your answer. The specific answer you obtain will depend on your choice of random number generator and seed. Show a segment of your box with randomly placed circles bounded by the square $(0, 0.025L)$ in each direction demonstrating that the circles do not overlap each other or the box edges. Marks will be awarded for correctness and efficiency [10], neatness/clarity of coding [5] and comments [5]. [20]
- (b) Modify your serial program to run in parallel using an appropriate parallelization strategy and method (eg OpenMP, MPI, CUDA, OpenCL, OpenACC). Justify your chosen parallelization strategy and method. Demonstrate how your program scales with different numbers of processes/threads and with problem size by considering different ratios r/L comparing both **strong** and **weak** scaling. Include appropriate graphs and clearly document all relevant parameters, including which machine(s) you have used, compilers and flags. The scaling tests should be conducted on HPC hardware such as Viking. Marks will be awarded on the basis of correct parallelization of the serial code [10], efficiency [5] and the scalability [15] of the code. [30]

End of paper